

Introduction to Matrices

Linear Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the order (dimensions) of a given matrix
- Locate specific entries in a matrix using row-column subscript notation
- Recognize square matrices and identify their main diagonals

Problems

1. State the order of the matrix shown below.

$$\begin{bmatrix} 4 & 7 & 2 \\ 1 & 9 & 6 \end{bmatrix}$$

2. State the order of the matrix shown below.

$$\begin{bmatrix} 5 \\ 3 \\ 8 \\ 2 \end{bmatrix}$$

3. Find the value of the entry $a(2,3)$ in the matrix shown below.

$$\begin{bmatrix} 3 & 0 & -1 & 4 \\ 7 & 2 & 9 & -5 \\ 6 & 1 & 8 & 3 \end{bmatrix}$$

4. Find the value of the entry $a(1,1)$ in the matrix shown below.



$$\begin{bmatrix} -4 & 6 & 2 \\ 5 & 0 & -3 \\ 1 & 8 & 7 \end{bmatrix}$$

5. Is the matrix shown below a square matrix? Explain your answer.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

6. Identify and list the entries on the main diagonal of the matrix shown below.

$$\begin{bmatrix} 6 & 3 & -1 \\ 2 & 9 & 4 \\ 7 & 0 & 5 \end{bmatrix}$$

7. Consider the matrix shown below. Determine whether entry $a(4,2)$ exists. If it does, state its value; if not, explain why.

$$\begin{bmatrix} 8 & -2 & 5 \\ 3 & 7 & 1 \\ 0 & 4 & -6 \end{bmatrix}$$

8. For the matrix shown below, find the values of $a(1,3)$, $a(2,1)$, and $a(3,4)$.

$$\begin{bmatrix} 2 & -1 & 7 & 0 \\ 4 & 3 & -5 & 9 \\ 6 & 8 & 1 & -2 \end{bmatrix}$$

9. Identify the order of the matrix below, list its main diagonal entries, and find the sum of those diagonal



$$\begin{bmatrix} 5 & 2 & -3 & 1 \\ 0 & -8 & 6 & 4 \\ 7 & 3 & 9 & -1 \\ -2 & 5 & 0 & 4 \end{bmatrix}$$

10. Matrix A has order 3×4 . Its entries are defined by the rule $a(i,j) = 2i - j$, where i is the row number and j is the column number. Write out the full matrix A, state whether it is a square matrix, and find $a(3,4)$.

$$a(i, j) = 2i - j$$



Introduction to Matrices — Answer Key

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Answer Key

1. Answer: 2×3 (2 rows, 3 columns)

- Count the number of rows: there are 2 rows.
- Count the number of columns: there are 3 columns.
- The order is written as rows $\times$ columns = 2×3 .

2. Answer: 4×1 (4 rows, 1 column)

- Count the number of rows: there are 4 rows.
- Count the number of columns: there is 1 column.
- The order is 4×1 .

3. Answer: 9

- The subscript (2,3) means row 2, column 3.
- Go to row 2: [7, 2, 9, -5].
- Move to column 3: the entry is 9.
- Therefore $a(2,3) = 9$.

4. Answer: -4

- The subscript (1,1) means row 1, column 1.
- Go to row 1: [-4, 6, 2].
- Move to column 1: the entry is -4.
- Therefore $a(1,1) = -4$.

5. Answer: Yes, it is a 3×3 square matrix (equal rows and columns).

- Count the rows: 3 rows.
- Count the columns: 3 columns.
- Since the number of rows equals the number of columns, this is a square matrix.
- It is a 3×3 square matrix.

6. Answer: 6, 9, 5

- The main diagonal consists of entries from the upper-left to the lower-right.
- Entry $a(1,1) = 6$ (row 1, column 1).
- Entry $a(2,2) = 9$ (row 2, column 2).
- Entry $a(3,3) = 5$ (row 3, column 3).
- The main diagonal is 6, 9, 5.

7. Answer: $a(4,2)$ does not exist; the matrix only has 3 rows.

- The matrix has 3 rows and 3 columns, so its order is 3×3 .
- The subscript (4,2) refers to row 4, column 2.
- Since there is no row 4 in a 3×3 matrix the entry is null (does not exist)



8. Answer: $a(1,3) = 7$, $a(2,1) = 4$, $a(3,4) = -2$

- $a(1,3)$: row 1, column 3 $\rightarrow$ entry is 7.
- $a(2,1)$: row 2, column 1 $\rightarrow$ entry is 4.
- $a(3,4)$: row 3, column 4 $\rightarrow$ entry is -2.

9. Answer: Order: 4×4 ; Main diagonal: 5, -8, 9, 4; Sum = 10

- Count rows: 4, count columns: 4 $\rightarrow$ order is 4×4 .
- It is a square matrix, so it has a main diagonal.
- Main diagonal entries: $a(1,1)=5$, $a(2,2)=-8$, $a(3,3)=9$, $a(4,4)=4$.
- Sum of main diagonal: $5 + (-8) + 9 + 4 = 10$.

10. Answer: Matrix A shown below; Not a square matrix; $a(3,4) = 2$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{bmatrix}$$

- Use the rule $a(i,j) = 2i - j$ for $i = 1,2,3$ and $j = 1,2,3,4$.
- Row 1 ($i=1$): $a(1,1)=2(1)-1=1$, $a(1,2)=2(1)-2=0$, $a(1,3)=2(1)-3=-1$, $a(1,4)=2(1)-4=-2$.
- Row 2 ($i=2$): $a(2,1)=2(2)-1=3$, $a(2,2)=2(2)-2=2$, $a(2,3)=2(2)-3=1$, $a(2,4)=2(2)-4=0$.
- Row 3 ($i=3$): $a(3,1)=2(3)-1=5$, $a(3,2)=2(3)-2=4$, $a(3,3)=2(3)-3=3$, $a(3,4)=2(3)-4=2$.
- The order is 3×4 ; since $3 \neq 4$, it is NOT a square matrix.
- $a(3,4) = 2$.

